EXPERIMENT 10

CO4 LAB-10

Program:

```
graph = {
    'A': [('B', 1), ('C', 3)],
    'B': [('D', 1)],
    'C': [('D', 1)],
    'D': []
}
h = {'A': 3, 'B': 2, 'C': 1, 'D': 0}
open_list = [(3, 'A')]
visited = []
while open_list:
    open_list.sort()
    f, node = open_list.pop(0)
    if node == 'D':
        print("Goal reached:", node)
        break
    if node not in visited:
        visited.append(node)
        for next_node, cost in graph[node]:
            g = cost + (f - h[node])
            open_list.append((g + h[next_node], next_node))
```

Output:

```
Goal reached: D
```